WEEK 4 · LESSON 19 OF 24 · HANDOUT

k-Means Clustering: Finding Customer Groups

Study time: **about 50 minutes** (reading, video and activity)

▶ **Watch:** https://www.youtube.com/watch?v=4b5d3muPQmA — open the lesson online to watch inside the course.

🎓 **Go deeper:** Harvard CS50 AI, Lecture 4: Learning (supervised, reinforcement, unsupervised) — https://www.youtube.com/watch?v=E4M_IQG0d9g

🎓 **Go deeper:** SciPy 2018 tutorial: Machine Learning with scikit-learn, Part 2 (scikit-learn core developers) — https://www.youtube.com/watch?v=gK43gtGh49o

Your goal in this lesson: Explain how k-means groups data without labels, choose the number of clusters with the elbow method and silhouette score, and describe each cluster in business terms.

Learning without answers

Until now every example had a label (rent, default). In **unsupervised learning** there is no answer column. The model looks for **structure** by itself. The most common task is **clustering**: putting similar examples into groups.

A mobile-money company has 500 customers and wants to design different offers. Nobody has labelled the customers — the company wants the data to suggest the groups.

```
import numpy as np
import pandas as pd

rng = np.random.default_rng(3)
def group(n, tx, amount, airtime):
    return np.column_stack([rng.normal(*tx, n), rng.normal(*amount, n), rng.normal(*airtime,
n)])

data = np.vstack([group(250, (8, 3), (4000, 1500), (0.6, 0.1)),
                  group(150, (45, 10), (15000, 4000), (0.15, 0.08)),
                  group(100, (20, 6), (60000, 15000), (0.05, 0.04))])
customers = pd.DataFrame(data, columns=["tx_per_month", "avg_amount_rwf", "airtime_share"])
customers["tx_per_month"] = customers["tx_per_month"].clip(1).round()
customers["avg_amount_rwf"] = customers["avg_amount_rwf"].clip(500).round(-2)
customers["airtime_share"] = customers["airtime_share"].clip(0, 1).round(2)
customers = customers.sample(frac=1, random_state=1).reset_index(drop=True)   # shuffle
print(customers.shape)
customers.head()
```

The three features are transactions per month, average transaction amount and the share of transactions spent on airtime.

How k-means works

1. Choose **k**, the number of clusters.
2. Place k **centroids** (centre points) at random.
3. **Assign** every customer to the nearest centroid.

4. **Move** each centroid to the average of its customers.
5. Repeat steps 3–4 until nothing changes.

Because the start is random, scikit-learn runs the whole thing several times (`n_init=10`) and keeps the best result. k-means uses **distances**, so scale first.

```
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans

X_scaled = StandardScaler().fit_transform(customers)
kmeans = KMeans(n_clusters=3, n_init=10, random_state=42)
customers["segment"] = kmeans.fit_predict(X_scaled)
print(customers["segment"].value_counts())
```

Describe the clusters

A cluster number means nothing until you describe it:

```
profile = customers.groupby("segment").mean().round(1)
print(profile)
```

You should find three clear groups:

Segment (your numbers may differ)	Behaviour	A name for the business
About 250 customers	~8 transactions a month, ~4,000 RWF each, mostly airtime	**Occasional airtime users**
About 150 customers	~44 transactions, ~15,000 RWF each	**Busy everyday users** (small traders)
About 100 customers	~20 transactions, ~60,000 RWF each, little airtime	**Big-ticket payers** (businesses, school fees)

Cluster numbers (0, 1, 2) are arbitrary — they can change between runs. The **descriptions** are what matter. Naming the clusters is a human job.

How many clusters? The elbow method

```
import matplotlib.pyplot as plt
from sklearn.metrics import silhouette_score

ks = range(2, 8)
inertias, silhouettes = [], []
for k in ks:
    km = KMeans(n_clusters=k, n_init=10, random_state=42).fit(X_scaled)
    inertias.append(km.inertia_)
    silhouettes.append(silhouette_score(X_scaled, km.labels_))

plt.plot(ks, inertias, marker="o")
plt.xlabel("k")
plt.ylabel("inertia (within-cluster spread)")
plt.title("Elbow method")
plt.show()
for k, s in zip(ks, silhouettes):
    print(k, round(s, 3))
```

- **Inertia** is the total squared distance from each point to its centroid. It always falls as k grows, so look for the **elbow** — where the drop suddenly slows. Here the elbow is clearly at **k = 3**.
- The **silhouette score** (−1 to 1) measures how well each point fits its own cluster compared with the next one. Higher is better. It peaks at **k = 3** (about **0.74**).

Always combine the numbers with **business sense**: 3 offers may be practical, 12 may not.

See the clusters

```
plt.scatter(customers["tx_per_month"], customers["avg_amount_rwf"],
            c=customers["segment"], s=12)
plt.xlabel("Transactions per month")
plt.ylabel("Average amount (RWF)")
plt.title("Customer segments")
plt.show()
```

Limits of k-means

- You must choose k.
- It finds **round**, similar-sized groups; it struggles with long, curved or very unequal clusters.
- Outliers pull centroids away.
- **DBSCAN** (density-based) finds clusters of any shape and marks outliers; **hierarchical clustering** builds a tree of clusters. Both are in `sklearn.cluster`.

Other unsupervised uses

- **Anomaly detection** — `IsolationForest` flags unusual transactions (possible fraud) without fraud labels.
- **Dimensionality reduction** — PCA, next lesson.

Key terms

- **Unsupervised learning** — finding structure in data without labels.
- **Clustering** — grouping similar examples.
- **Centroid** — the centre of a cluster.
- **Inertia** — within-cluster squared distances.
- **Elbow method / silhouette score** — ways to choose k.

Remember this: k-means groups similar customers around centroids; choose k with the elbow, the silhouette score and business sense, then describe each group in plain words.

Name: Date:

Activity

Task 1

Run k-means with k = 4. Which segment splits in two? Is the split useful for the business?

Task 2

Run k-means on the **unscaled** data. Compare the segment sizes with the scaled result and explain the difference.

Task 3

Use `IsolationForest(random_state=42)` on the scaled data and look at the 10 most unusual customers.

Task 4

Write one marketing idea for each of the three segments.

Self-check

Think about your answers here, then take the quiz online — it is marked instantly and explains every answer.

1. What makes clustering unsupervised?
   - A. It predicts rent
   - B. It uses no data
   - C. There are no labels; the model finds groups by itself
   - D. It needs a test set
2. In k-means, what happens after customers are assigned to the nearest centroid?
   - A. The model stops
   - B. Each centroid moves to the average of its customers
   - C. Labels are added
   - D. Features are deleted
3. Why scale features before k-means?
   - A. Distances would be dominated by the feature with the biggest numbers
   - B. To add labels
   - C. k-means requires text
   - D. It is not needed
4. What does the elbow method look for?
   - A. The largest cluster
   - B. The k where inertia stops falling quickly
   - C. The highest inertia
   - D. k = 1
5. A silhouette score close to 1 means…
   - A. The model failed
   - B. Clusters overlap badly
   - C. Points fit their own cluster well
   - D. There is one cluster
6. Which method finds clusters of any shape and marks outliers?
   - A. Naive Bayes
   - B. DBSCAN
   - C. Ridge
   - D. Linear regression

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